BIS

Invitation to Tender (ITT)

Project Kairos (hereinafter called "the project")

Dispatch Date:	**02.03.2026**
Acknowledgment of participation in ITT & Deadline for queries from bidders ("Query Date"):	[09.03.2026] by e-mail to corporate.procurement@bis.org & han-chih.yang@bis.org
Deadline for answers to be sent to all bidders by email	16.03.2026
Deadline for offer submission ("Submission Date"):	27.03.2026
Period for possible presentations at BIS	13.04.2026 - 17.04.2026
Expected date of decision by the BIS:	18.05.2026
Envisaged start of contract:	15.06.2026

Requestor Contact (the "Requestor") han-chih.yang@bis.org

Confidentiality

All information contained herein is to be treated confidentially. It is to be used solely for the purpose of drawing up a quote. Please refer to section 5.3 ("Obligation of confidentiality").

Contents

1. Introduction and Overview

1.1 The Bank for International Settlements

The Bank for International Settlements (BIS) is an international organisation with its head office at Centralbahnplatz 2, CH-4051 Basel, Switzerland, and representative offices in Mexico and Hong Kong SAR. The BIS's objectives are to promote cooperation between central banks, to provide a centre for economic and monetary research, and to act as a bank for central banks More information about the BIS can be found on our website: https://www.bis.org.

As an international organisation serving central banks in pursuit of monetary and financial stability, the BIS recognises the increasing impact that technology-driven innovation will have on global financial systems and on the way central banks conduct their core public missions. In light of this, the BIS has established an Innovation Hub to serve as a dedicated platform for acquiring expertise and developing technology and digital innovation projects for the benefit of the central banking community. The BIS Innovation Hub (BISIH), which is part of the BIS itself without separate legal personality, currently has regional Hub centres in Frankfurt/Paris (Eurosystem Centre), Hong Kong SAR, Singapore, Switzerland, London, Stockholm (Nordic Centre) and Toronto.

The mandate of the BISIH is to identify and develop in-depth insights into critical trends in financial technology of relevance to central banks, to explore the development of public goods to enhance the functioning of the global financial system, and to serve as a focal point for a network of central bank experts on innovation. It complements the already well-established cooperation within the BIS-hosted committees.

The BISIH was set up given that:

- technology-driven innovation in financial services is accelerating, and will have major consequences for global financial systems and central banks;
- the IT revolution has repercussions in multiple locations simultaneously;
- central banks can achieve economies of scale by working together.

2. Scope and Requirements of the ITT

2.1 Project summary

The project seeks to study and understand the possible design options available to central banks to foster interoperability via token and bridge design, while maintaining adequate controls over their reserves. For example, central banks will need to restrict access to tokenised reserves; manage how these tokens can be used (e.g. limits) and define which ledgers they can be used on. Similarly, central banks may want to ensure operational and technical controls on tokenised assets mobilised as collateral on ledgers not operated by them. The project does not seek to argue in favour of a particular ledger type or ecosystem design.

The project will deliver a multi-ledger testing environment designed to explore and validate how tokenised assets can move across different blockchains networks. The environment will bring together:

- Multiple DLT ledgers – spanning EVM and non-EVM architectures, across both public permissionless and private permissioned ledgers. Each ledger serves a distinct role in the token lifecycle

- Cross-chain bridges – enabling tokens to move securely between ledgers with varying technical architectures. Designed to preserve integrity and ensure consistency across different trust models

- Programmable token instruments – covering three token types: reserves, deposits and securities. Each token type carries distinct programmable rules, from access controls and transfer restrictions to automated lifecycle events

- Transaction orchestration – coordinating complex, multi-step operations across ledgers, from burns and mints to transfers and settlements. Manages the sequencing and dependencies throughout the process to support controlled and complete execution

- Control and visualisation layer – allowing users to configure, trigger and monitor orchestrated transactions. This includes a clear view of token movements, ledger states and transaction flows as they progress

Subject to the execution of the contract between the BIS and the selected bidder, the BIS shall become the exclusive owner of intellectual property rights and any other proprietary rights of all deliverables, including, but not limited to materials, standards, concepts, files, software or technology, developed or produced by the selected bidder, either on its own or in collaboration with the BIS, as part of the contract performance. The selected bidder shall take all necessary steps, execute all necessary documents and generally assist in securing such proprietary rights and transferring or licensing them to the BIS in compliance with the requirements of the applicable law.

This arrangement is intended to ensure that the outputs of the project can be used, shared, and built upon by the BIS and other project partners for public policy, research, and future collaborative initiatives, without dependency on proprietary bidder constraints.

With respect to pre-existing intellectual property rights relating to the deliverables, the BIS will receive a non-exclusive, transferable right of use without restrictions in terms of time, space, and substance, which grants the BIS the possibility to use and dispose of the deliverables. The selected bidder will undertake not to establish any rights based on those pre-existing intellectual property rights which could be asserted against the possibilities of use granted herein. In particular, it undertakes to transfer or license these intellectual property rights only subject to the rights of use of the BIS.

2.2 Project overview (incl. requirements and scope)

For a detailed description of the requirements and scope, please refer to statement of work in Annex 2. It is expected that the project be delivered in phases:

Phase 1 – Building the experimentation sandbox:

- The sandbox will be created and hosted within the BISIH cloud environment which provides security and access controls. Key activities in this phase include:

- Provision and configure DLT test environments across both permissioned and permissionless ledgers. Depending on the ledger, this may include local installation and configuration of nodes within the sandbox environment, establishing connectivity to public testnet environments, or a combination of both. All environments must support the deployment of smart contracts and token standards required for the testing phases.

- Any smart contracts, configurations, and metadata deployed on public networks and environments must be fully redacted to exclude any identifiable information related to the project, including its personnel and organisations (stakeholders, partners, vendors, contractors, institutions or central banks), project and system identifiers, and any data that could be used to infer the identity of any participant or affiliated entity. The vendor is responsible for monitoring and correcting any inadvertent exposure. If full redaction cannot be achieved, public networks and environments shall not be used.

- Adapting existing emulations of a real-time gross settlement (RTGS) and a Central Securities Depository (CSD) system for use within the sandbox environment.

- Design and develop tokenised reserve, tokenised deposit and tokenised securities, with implementations on permissioned and permissionless ledgers as required.

- Develop cross-ledger bridge mechanisms to enable the transfer of tokens between ledgers with diverse technical standards. Bridge designs must support both centralised models, operated by a trusted intermediary, and decentralised models, using consensus-based validation

Phase 2 – Testing use cases

- **Phase 2a.** – Testing simple liquidity transfers and control functionality (See use case 1).

- **Phase 2b.** – Testing complex transfers across multiple ledgers and advanced programmability (See use case 2 and 3).

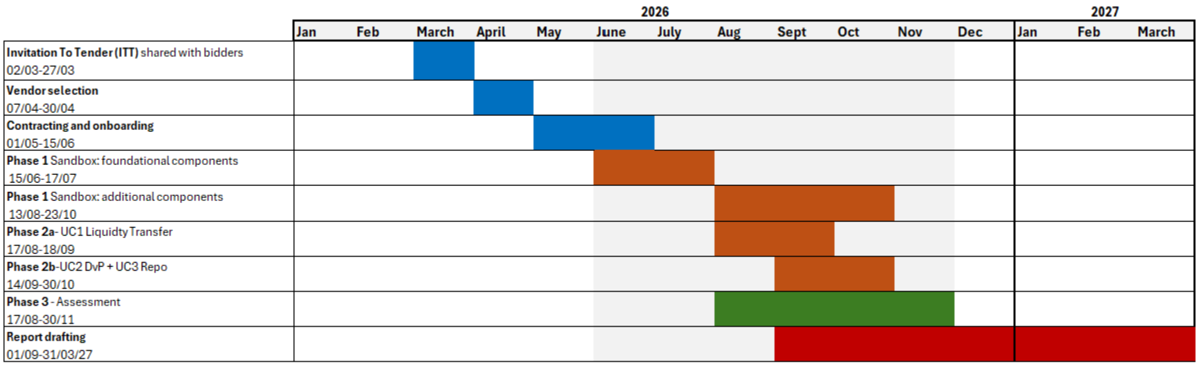
Phase 3 – Assessment:

- The results of the testing in Phase 2 will be examined through an assessment framework that will consider the security, resilience, scalability and efficiency of the transactions. The assessment will be created in conjunction with the selected bidder and will include technical documentation, a comprehensive analysis of selected designs and a comparison of other models not implemented, all of which will be referenced in the technical annex of the project report. It is imperative that delivery of phases 1, 2a and 2b is completed by end of October 2026. Delivery of these phases is directly linked to phase 3 assessment, which runs in parallel, and must have concluded by the end of November 2026

Report Drafting

- Completion of these phasesand the associated findings will be used for a project report scheduled for completion by March 2027 (see below timeline). A core requirement for the completion of the project report is the assistance from the selected bidder to work with the project team.

2.3 Project key dates



2.4 Project team

The project team will be comprised of members of the BIS Innovation Hub and project partners. This will include but not be limited to, a project manager, technical lead, and business analyst, as well as access to subject matter experts to guide the business requirements.

The selected bidder is expected to provide project management and business analysis support along with technical expertise and delivery. The vendor must propose their preferred team(s) structure for delivery. To support core delivery between June-November, it is advisable that the selected vendor has the resource capability and capacity to run a two team structure, as a minimum.

In the event that the bidder proposes a consortium bid for this project, the bidder must provide detailed information regarding the consortium arrangement and clearly explain the rationale for such a proposal. Only one contract will be established with the main bidder, who will be fully responsible for ensuring that all subcontractors are properly instructed, comply with all obligations outlined in the contract, and do not transfer any contractual obligations to third parties without the prior written consent of the BIS. Disclosure of confidential information to subcontractors or other third parties is permitted only on a need-to-know basis and solely where the recipient: (i) requires the information for the project; (ii) has been informed of its confidential nature; and (iii) is bound by written confidentiality obligations no less stringent than those applicable to the main bidder.

2.5 Project use cases

The project will primarily focus on testing three use cases.

Use case 1: Bridging tokenised reserves between two private permissioned ledgers (EVM to non-EVM)

A commercial bank wishes to move its own tokenised reserves between two permissioned ledgers, EVM and non-EVM. The bridge therefore needs to orchestrate a transfer of tokenised reserves from one to another:

- Burn the tokenised reserves from commercial bank A's wallet.
- Mint corresponding tokenised reserves to commercial bank A's wallet on the other Network.

Key learnings – This use case will provide the following learnings:

- How could tokenised reserves be bridged between two permissioned DLTs with very different technical architectures?
- How do we validate that funds have been created and destroyed to avoid double-spend?
- How do we ensure that the bridge itself is resilient to attacks? How do we minimise third-party risks to build resilience into the system?
- How DLT bridges could be designed to introduce appropriate safeguards to support the transfer of tokenised reserves across different types of DLT ledgers. What are the benefits of centralised vs. decentralised bridges?
- How could programmable logic be embedded in reserve tokens (access controls, ability to freeze and repatriate) and maintained after a bridging operation?

Use case 2 – DvP transaction on a public permissionless ledger (EVM), using tokenised reserves on a private permissioned (EVM) DLT

The second scenario focuses on settling a tokenised DvP transaction in a multi-ledger ecosystem. A customer of Bank A is seeking to purchase a tokenised bond from a customer of Bank B on permissionless ledger using tokenised deposits.

The bond issuer's bank (Bank B) will need to convert the funds into Bank B's deposit tokens. Both banks need to settle the resulting obligation in tokenised reserves on a permissioned ledger being operated by a consortium. In this scenario, a bridge would be used to:

- Burn Bank A's deposit tokens from Bank A's customer's account on permissionless ledger.

- Instruct the transfer of tokenised reserves from Bank A to Bank B on permissioned ledger.
- Mint Bank B's tokens on permissioned ledger in Customer B's account and transfer the tokenised security to Customer A.

  Ensure all these steps happen atomically (i.e. all or nothing).

Key learnings – This use case will provide the following learnings:

- How to orchestrate a tokenised DvP transaction on a public permissionless ledger that is settled using tokenised reserves held on a private permissionless ledger.
- How to link these movements to happen atomically across different DLT ledgers.
- How to design the trust and authority model to allow a DLT bridge to burn and mint tokenised assets with appropriate safeguards.

Use case 3 – Repurchase agreement on a private EVM permissioned ledger using collateral from a public non-EVM permissionless ledger

In the third example, a commercial bank would like to borrow tokenised reserves from a central bank's repo facility to plug a liquidity shortfall. They have eligible collateral in public non-EVM permissionless ledger while the repurchase facility is operating on a permissioned EVM ledger. Collateral is therefore mobilised from a public permissionless ledger onto the permissioned ledger for use in the repurchase facility.

A bridge is used to move a tokenised security from public non-EVM permissionless ledger to private EVM permissioned ledger to be used as collateral in a repo transaction. In this scenario, a bridge would be used to:

- Transfer eligible collateral (a tokenised security) from public non-EVM permissionless ledger to private EVM permissioned ledger.
- Enter a repurchase agreement with the central bank to provide reserves to the commercial bank's wallet on private EVM permissioned ledger.
- At the agreed time, unwind the repo on EVM permissioned ledger and brings back the tokenised security to the original ledger.

Key learnings – This use case will provide the following learnings:

- How to bridge tokenised securities across two ledgers (permissioned and permissionless).
- How to maintain programmable logic in the securities token (coupon payments, etc.).
- How to successfully reverse the transaction, repaying the repo and repatriating the asset.
- How to design bridges to introduce appropriate safeguards to support the settlement of tokenised securities.

Comparative analysis of settlement model: using tokenised reserves versus reserves in RTGS (Under use cases 2 and 3)

- Use case 2: a private EVM permissioned ledger could be replaced with an orchestrated transaction in the RTGS system. The RTGS will enable the reservation of reserves, and subsequent settlement of those reserves, to ensure the use case is atomic. This comparison will allow the central banking community to understand the relative trade-offs of DLT native settlement vis-a-vis synchronised settlement in the RTGS system.

- Use case 3: a private EVM permissioned ledger movement of tokenised reserves will be replaced by a RTGS system. Private EVM permissioned ledger will still be used to hold the tokenised security, and the repurchase agreement smart contract. This comparative use case can build on existing synchronised RTGS settlement and a tokenised CSD to orchestrate repurchase agreements. This comparative analysis will allow us to assess the benefits of DLT-native repo transaction relative to synchronising.

Existing components and resources available to the bidder

The selected bidder is required to utilise BISIH cloud infrastructure for all project-related work. The following existing components and infrastructure within the BISIH will be provided for use during the project:

- Emulated versions of the Real-Time Gross Settlement (RTGS) and Central Securities Depository (CSD) systems
- A dedicated sandbox environment in the BISIH Azure Cloud

The vendor shall not provision or use any external infrastructure, networks, or services outside of BISIH environment for project-related work without prior written approval.

3. Instructions to bidders

3.1 Scope of quote

The bidder's quote shall be submitted in accordance with the scope and requirements of this ITT as set out in section 2 above.

3.2 Project reference

Any document, letter or e-mail from the bidder should quote the reference ITT "Project Kairos".

3.3 Confirmation of participation

Please acknowledge your participation in the tender on or before the date indicated on page 2.

When confirming your participation, please indicate a contact person (name, address, telephone number, e-mail address) and availability for either an onsite or virtual presentation(s)

3.4 Validity of the quote

All quotations submitted in response to this ITT shall remain valid for a period of six (6) months from the Submission Date. A quote valid for a shorter period shall be rejected by the BIS as non-responsive to the ITT requirements. In exceptional circumstances and prior to the expiration of the validity of the quotes, the BIS may request bidders to extend the period of validity of their quotes providing the background to such request. The request and responses will be made in writing. A bidder may refuse the request. A bidder accepting the request for extension shall not be permitted to modify their quote in any way.

3.5 Language and format

Bidders should submit their quote and all accompanying documents in English to corporate.procurement@bis.org and han-chih.yang@bis.org . The maximum size per e-mail is 25 MB.

Any correspondence related to this ITT shall be in plain English.

3.6 Requests for clarification from bidders

Any requests for clarifications of this ITT should be submitted in accordance with the deadlines set out on page 2 of this document, and solely via e-mail to the Requestor.

The Requestor is the sole contact at the BIS for the bidder regarding this ITT. Queries regarding this ITT may not be addressed to other contacts inside or outside the BIS without prior written approval of the Requestor. Non-compliance with this rule may lead to disqualification from the ITT.

The query, as well as the BIS's response, in writing, will be notified to all bidders at the same time, without identifying the bidder who initiated the query.

3.7 Modifications by the BIS

The BIS reserves the right to amend the information and requirements contained in this ITT at any time prior to the Submission Date and for any reason (incl. in response to a clarification requested by prospective bidders), without any liability to any bidder for costs, losses, or expenses incurred in connection with the preparation or submission of a tender or participation in this process. Any amendment issued shall form part of this ITT and shall be communicated in writing to all bidders that have received this ITT document. If the BIS deems it necessary (e.g. because the amendment to the ITT requirements would likely lead to delays), it will extend the Submission Date and inform all bidders accordingly.

3.8 Invitation to make a presentation

Prior to accepting an offer, the BIS may invite the bidder to make a virtual and/or in-person presentation at the Bank, as part of the ITT procedure and at no additional cost to the BIS.

In order to assist the examination, evaluation and comparison of offers and qualification of bidders, the BIS may also, at its discretion, verify any information provided in the offer and/or ask any bidder for clarifications of its offer, allowing reasonable time for response. Any clarification submitted by a bidder that is not in response to a request by the BIS shall not be considered. The BIS's request for the clarification and the bidder's response shall be in writing.

3.9 Right to terminate the ITT procedure

The BIS reserves the right to terminate the ITT procedure at any time prior to the award of the contract, without any liability to any bidder for costs, losses, or expenses incurred in connection with the preparation or submission of a tender or participation in this process.

3.10 Bidder's right to change/withdraw an offer, BIS's right to accept or reject an offer

Bidders have the right to change or withdraw an offer up until the Submission Date.

The BIS is entitled to reject any offer at its sole discretion and without any explanation, and without incurring any liability to the affected bidder(s). Nothing in this ITT shall oblige the BIS to award a contract.

3.11 Costs arising from participating in the ITT procedure

The BIS shall not be responsible for, or pay any costs or expenses which may be incurred by the bidder for participating in the ITT, including preparation and submission of their offer, attendance at any on-site visits, presentations etc.

3.12 Correction of arithmetical errors

If a substantially responsive quote contains arithmetical errors, the BIS shall correct these on the following basis:

1. Discrepancy between the amounts in the price breakdown and the amount given under the Total Price – the amounts in the price breakdown will prevail;
2. Discrepancy between the unit price and the total price that is obtained by multiplying the unit price and quantity - the unit price shall prevail, unless there is an obvious misplacement of the decimal point in the unit price, in which case the total price as quoted shall govern.

4. Content and signature of the quote

Bidder are asked to structure their offer as follows:

1. The bidder is requested to structure their response within the following page limits: Company Overview and General Information (up to 5 pages), Technology and Architecture Approach (up to 3 pages), Key Personnel (up to 3 pages)
    a. The Technology and Architecture section should include a single-page conceptual diagram with a supporting narrative explaining how the bidder envisages delivering the solution
2. General information about the bidder, organisational set-up, proposed team structure(s) and size(s),
3. Description of the services to be provided, including project management and timeline, etc.
4. Price information. VAT (if applicable) should be shown as a separate line item Delivery dates and timelines, aligned to project milestones, including key dependencies and assumptions broken down by month.
5. Confirmation of the BIS retains sole and exclusive ownership of all intellectual property rights for all the deliverables.

4.1 General information about the bidder and organisational set-up

- A company overview, including size of firm and a diagram of the overall organisation, figures for the turnover of the last two fiscal years and a projection for the next year.
- A description of the company's core business, global presence and its strategy.
- A detailed description of the internal organisation of the unit that will provide the required services.
- An overview of key risks, their likelihood and impact, and proposed mitigations
- Any proposed deviations or adjustments to the project milestones and timelines provided
- At least two case studies/references of where the bidder delivered the same or similar services as requested in this ITT.
- For companies in Switzerland, an extract from the commercial register ("Handelsregister-auszug") showing the authorised signatories, and a statement from the debt enforcement office ("Betreibungsauskunft") not older than three months. For bidders located outside Switzerland, similar documents used in local commercial practice.

4.2 Description of proposed services

- The offer should be substantially compliant with all the requirements set out in sections 2 and 3 of this ITT. Provided that an offer is substantially compliant, the BIS may request the bidder to promptly submit any necessary missing information or documentation, to rectify non-material omissions in the offer or for verification of information/documentation submitted.

- Please provide names, titles and detailed qualifications including relevant experiences with the ledgers and technologies in scope. (e.g. curriculum vitae) of the bidder's personnel that would be assigned to the project.
- Please detail what backup solution(s) you have in place for the staff providing the services.
- Please specify if you would be using subcontractors for the fulfilment of the services (if so, please give details). Please also explain the reason for using subcontractors.

4.3 Price information

All prices and rates shall be quoted in British Pounds Sterling (GBP), Swiss Francs (CHF) or United States Dollars (USD) and will be converted to CHF Swiss Francs (CHF) for evaluation purposes. VAT is not applicable.

1. Fixed-Price Package

   Bidders are required to submit a fixed-price package for the project and each team (if multiple teams are proposed) as part of their proposal. The submission must include a comprehensive cost breakdown linked to a milestone-based delivery model. The cost breakdown should clearly detail the following:

   - Assigned roles and their respective seniority levels.
   - Estimated number of workdays required.
   - Applicable daily rates and any discounts offered.

2. Rate Card for Ongoing Support

   Bidders must also provide a rate card for ongoing support during report writing phase, until March 2027.

The prices submitted in the quote are considered to be final and they will therefore not be subject to a round of competitive renegotiation amongst the bidders. If required, a clarification of the costs and its assumptions can be made.

The BIS enjoys a special tax status as an international organisation in United Kingdom, therefore VAT is exempt

4.4 Signature

An offer submitted to the BIS pursuant to this ITT shall be signed by a duly authorised representative of the bidder so as to be legally binding on the bidder, in the event of an award of contract.

5. General terms and conditions for this ITT

5.1 BIS contractual documentation

Any award under this ITT shall be binding on the BIS only subject to the conclusion of a written contract between the BIS and the bidder. Any contract concluded based on this ITT shall be governed by and construed under the laws of Switzerland, and in the event of a dispute, Basel-Stadt/CH shall be the exclusive place of jurisdiction.

Submission of a quote constitutes acceptance of the attached BIS contractual documents, without any changes (see Annex 1).

The BIS will not accept any terms and conditions of a bidder which conflict with the terms and conditions of the BIS and/or its standard agreements. If the ITT scope requires a subscription or software license, please add these specific terms in addition to the acceptance of the BIS Terms and Conditions.

The BIS retains the right to verify all information supplied by the bidder and request any further supporting documentation it deems necessary.

The envisaged start date of the contract is 15.06.2026.

5.2 Conflict of interest

The bidder warrants that its participation in this ITT and contractual performance, if awarded the contract pursuant to this ITT, would not give rise to any actual or potential conflict of interest. The bidder must immediately inform the Requestor of any potential conflict, including if they are in doubt as to whether a conflict of interest exists. In all cases, the Bank's determination as to whether a conflict of interest exists shall be final.

In addition, the bidder warrants that it, its directors, employees, agents, subcontractors, suppliers and any other affiliated entities (natural or legal) shall observe the highest standards of transparency and integrity during the procurement, execution and implementation of any contract awarded pursuant to this ITT.

The bidder further warrants that it, its directors, employees, agents, subcontractors, suppliers and any other affiliated entities (natural or legal) have not and will not engage in any corrupt, coercive, collusive, fraudulent or obstructive practices, theft or misuse of the BIS resources.

Failure to meet any of the requirements and warranties under this section will result in immediate disqualification of a bidder, or cancellation of award and/or termination of contract and indemnification to the BIS, if the failure to meet such requirements and warranties is discovered after the award or conclusion of the contract.

The BIS retains the right to request the signing of a conflict of interest statement when deemed necessary.

5.3 Obligation of confidentiality

This ITT and all information relating to the project constitute Confidential Information under the non-disclosure agreement entered into between the bidders and the BIS, which remains in full force and effect.

The BIS will treat as confidential all information relating to the submitted quote.

6. Evaluation criteria and process

6.1 Evaluation criteria

The submitted offers will be evaluated according to the following criteria:

Criteria	Weight
Appropriateness of the skill set and experience of the bidder	25%
Approach to core objectives and project phases	25%
Development teams and timeline alignment	15%
Pricing	15%
Quality of the offer (completeness, accuracy)	15%
Acceptance of BIS contractual documentation	5%
Total	100%

The evaluation process consists of the following steps:

- After the submission of the offer the BIS will analyse and evaluate the documents received.
- After the evaluation the BIS may shortlist selected bidders.
- Shortlisted bidders may be asked to present their offers (see section 3.8 above).

Based on the outcome of the evaluation process, the BIS and relevant committees will confer next steps and may select one bidder for delivery of the services set out in this ITT.

Based on the outcome of these activities, the Bank will decide on the next steps and may select one bidder for delivery of the services.

7. Annexes

Annex 1a: BIS Terms and Conditions for IT Services

Annex 1b: BIS IT Services Contract Template

Annex 2: Project Kairos: Statement of Work